START HERE · 00–08 MINUTES

Pure Mathematics

Thursday 5 November 2026 · 09:00–10:00 · PRIORY RD CMPLX F BLK 3F11

Key ideas before the questions

For $f : X \to Y$: **injective** means $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; **surjective** means every $y \in Y$ has some $x \in X$ with $f(x) = y$. Bijective means both.

For $A \subseteq X,\ B \subseteq Y$:

$$f[A] = \{f(x) : x \in A\}, \qquad f^{-1}[B] = \{x \in X : f(x) \in B\}.$$

An image follows inputs forward. A pre-image collects all inputs landing in the target set, even when no inverse function exists.

1 · Classify the maps

For each map, decide whether it is injective, surjective, bijective, or neither. Give one definition-based reason.

Map	Classification	Reason
$f : \mathbb{R} \to \mathbb{R},\ f(x) = 2x - 3$		
$g : \mathbb{R} \to \mathbb{R},\ g(x) = x^2$		
$h : \mathbb{R} \to [0, \infty),\ h(x) = x^2$		

2 · Image and pre-image

Let $p : \mathbb{R} \to \mathbb{R},\ p(x) = x^2 - 1$. Find $p([-1, 2])$ and $p^{-1}([0, 3])$. Then explain why these are different kinds of sets.

Image

Pre-image

Why the two answers differ

Definitions and proof structure

Continue the Pure Mathematics worksheet

> **Proof checklist**
>
> For injectivity, start with equal outputs. For surjectivity, choose an arbitrary target in the codomain and construct a valid input. For a counterexample, state the domain, codomain, and the failed property.

3 · Work with definitions

Let A, B be sets and $f : A \to B$.

1. Write the definitions of injective and surjective in words and symbols.
2. If f and g are injective, prove that $g \circ f$ is injective.
3. Give a concrete function that is surjective but not injective. Explain exactly where injectivity fails.

Definitions

Composition proof

Concrete example and failure

4 · PS5 peer feedback

Choose two peer submissions. For each one, record one clear step, one step needing a stronger definition or justification, and one question for the author.

Submission	Clear step	Improvement / question
A		
B		

Exit check

What changes when the codomain of x^2 changes from $\mathbb{R}$ to $[0, \infty)$?

One precise sentence

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Linear Algebra

Friday 6 November 2026 · 09:00–10:00 · FRY BLDG G.14

Key ideas before the questions

Span means all linear combinations. A **basis** is linearly independent and spanning. Its size is the dimension.

For subspaces,

$$\dim(X+Y) = \dim X + \dim Y - \dim(X \cap Y)$$

.

Row reduction reveals pivots, rank, free variables and the null space. For $T : \mathbb{R}^n \to \mathbb{R}^m$, $\dim(\ker T) + \dim(\operatorname{im} T) = n$.

1 · A subspace with a complex scalar field

Let $V = \{(x, y, z, x + iy) \in \mathbb{C}^4 : x, y, z \in \mathbb{C}\}$. Show that V is a complex subspace of $\mathbb{C}^4$. Find a basis and $\dim V$.

Subspace check

Basis and dimension

2 · Sum and intersection

Let $X, Y \subseteq \mathbb{R}^n$ be subspaces with $\dim X = m$ and $\dim Y = k$.

1. Is $\dim(X + Y) = m + k$ always true?
2. State and justify the correct formula.
3. Give one example with trivial intersection and one without.

Answer and formula

Examples

Independence, null spaces and rank

Continue the Linear Algebra worksheet

3 · Independence and basis

Decide whether the following vectors in $\mathbb{R}^3$ are linearly independent:

$$v_1 = (1, 2, -1), \quad v_2 = (1, 1, 1), \quad v_3 = (0, 1, -2).$$

Use a homogeneous system or row reduction. Do they form a basis of $\mathbb{R}^3$?

Set up the homogeneous system

Row reduction and conclusion

4 · Null space and rank

Find the null space and rank of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 0 \end{pmatrix}$$

Give a basis for the null space and explain how the number of pivots gives the rank.

Row-reduced form

Null-space basis and rank

5 · Rank–nullity

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be $T(x, y, z) = (x + y, y + z)$. Find $\ker T$, $\operatorname{im} T$, rank and nullity. Check rank–nullity.

Kernel and image

Rank–nullity check

CLOSE · 25–30 MINUTES

Final self-check

Use this page to prepare one question for the tutorial

Before class

Can I explain why a proposed basis both spans and is independent? Can I identify pivots and free variables after row reduction?

Choose one question from the worksheet and write a complete solution another student could check line by line.

Question number and first step

Complete solution outline

One question to bring to class

My question